PMC: Build-Time Per-Modality Centroid Correction for Cross-Modal Binary-Quantized Retrieval

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Abstract

Approximate nearest neighbor search is a core operator in large-scale retrieval, where vector quantization is widely used to reduce memory cost. Binary quantization methods encode each dimension as a single bit, but they assume queries and database vectors share a common centroid. This assumption fails in cross-modal retrieval, where each modality clusters around a different centroid, producing a modality gap that corrupts the centroid's dual role as inverted-file routing pivot and sign-bit decision boundary. We propose Per-Modality Centroid Correction (PMC), which shifts database vectors toward the query centroid at build time, rewriting routing and code boundaries at the source rather than merely adjusting queries at serving time. A selective variant confirms that concentrated gap energy in a few dimensions drives most recall loss. PMC adds no index memory or serving overhead; at α=1, the correction is fully absorbed into the index with zero query-time cost. Experiments on five cross-modal benchmarks, three encoders, and four binary-quantized index types show gains over uncorrected and query-shifted baselines that scale with modality-gap magnitude, up to a 400M-vector deployment. Code: https://github.com/sehoon787/PMC.

CCS Concepts

• **Information systems → Multimedia and multimodal retrieval**; *Search engine indexing*.

Keywords

cross-modal retrieval, modality gap, approximate nearest neighbor search, binary quantization

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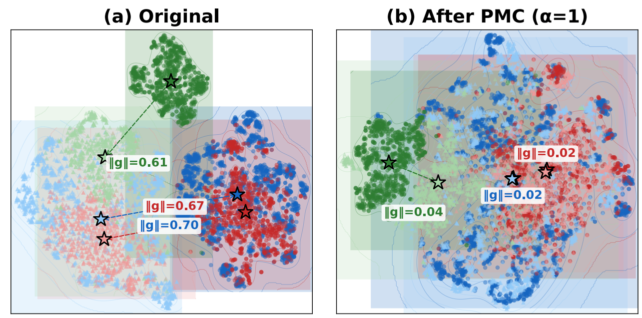

Figure 1: t-SNE of ImageBind embeddings. (a) Each modality forms a separate cluster. (b) After PMC (α=1), paired modalities overlap. Blue = MSCOCO, red = Flickr30K, green = Clotho; dark = DB, light = query.

1 Introduction

Multimodal foundation models such as CLIP [23] and ImageBind [9] map heterogeneous signals (images, text, audio) into a shared vector space for cross-modal vector similarity search [15]. At deployment scale, storage becomes the bottleneck: high-dimensional float32 embeddings can require terabytes of RAM, motivating approximate nearest neighbor (ANN) indexes that compress to a few bytes per vector [7, 14].

Binary quantization (BQ) is the extreme compression point: RaBitQ [7] reaches 32× compression by encoding each dimension as one sign bit relative to the dataset centroid. Its adoption in Apache Lucene BBQ and multi-bit Extended-RaBitQ [6] shows BQ indexing is now deployed in production.

These methods rely on a shared-centroid residual premise: queries and database vectors share a common centroid. Cross-modal retrieval violates this assumption—queries from one modality cluster around a different centroid than the database vectors.

This *modality gap* [21] is particularly harmful to BQ because the encoding $\text{sign}(x_i - c_i)$ has *zero error margin*: the decision boundary passes exactly through the centroid, and any systematic displacement causes structured bit corruption that scales with gap magnitude (Figure 1); an oracle test (MSCOCO, CLIP-L) confirms this: same-modality queries ($\|g\|$=0) reach R@100=0.71 under RaBitQ, versus 0.54 for cross-modal queries ($\|g\|$=0.82). Prior work addresses cross-modal retrieval through graph augmentation [2, 13], encoder fine-tuning [5], or embedding-space calibration [19], but none targets compressed ANN indexes where centroid misalignment corrupts both Inverted File (IVF) routing and binary codes.

In this paper, we first analyze how the modality gap degrades BQ ANN indexes at two coupled stages—IVF routing and binary code formation—and show that dimensions with concentrated gap

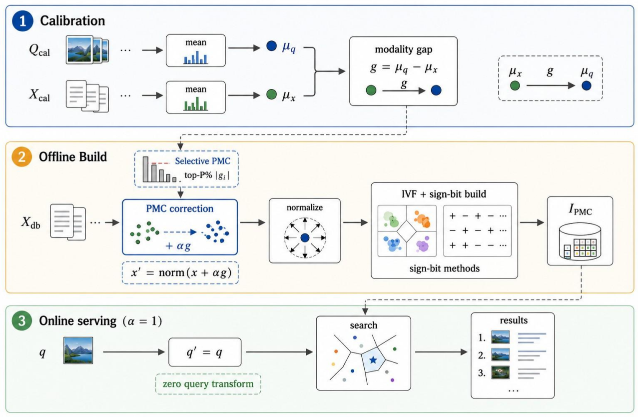

Figure 2: PMC workflow: calibration estimates μ_q, μ_x, and g=$\mu_q - \mu_x$; offline build shifts/normalizes database vectors before BQ index construction; at α=1, online serving uses q′=q and searches $\mathcal{I}_{\mathrm{PMC}}$.

energy are most vulnerable to boundary crossing; a selective correction test confirms that correcting only the highest-gap dimensions recovers most of the gain. Based on this analysis, we propose Per-Modality Centroid Correction (PMC), a one-time build-time alignment that shifts database vectors toward the query centroid before index construction. Unlike query-only mean shift, which leaves IVF centroids and quantized codes misaligned, PMC exploits the centroid's dual role as IVF routing pivot and sign-bit decision boundary, rewriting both at the source with no serving-time query transform and no additional index memory (Figure 2). We validate PMC across five cross-modal benchmarks including 407M-scale LAION, three multimodal encoders, and four BQ index types. Our contributions:

- We show that the modality gap corrupts both IVF routing and binary codes, with recall loss scaling with gap magnitude.
- We discover that this corruption concentrates in a few high-$|g_i|$ dimensions; a selective PMC variant correcting only those recovers most recall.
- We propose PMC, a zero-cost build-time correction (α=1) that realigns routing and codes via the centroid's dual role; ablation confirms DB-side over query-side correction.

2 Related Work

Vector quantization for ANN. Classical ANN compression methods such as Product Quantization (PQ)/OPQ [8, 14] quantize vectors by subspace codebooks, while newer systems such as ScaNN/SOAR/QAQ [12, 28, 34] improve scoring, routing, or query-aware ranking through anisotropic quantization and query-dependent computation. In the 1-bit regime, RaBitQ [7] provides distortion bounds for BQ, Extended-RaBitQ [6] generalizes this line to multi-bit settings, and later variants integrate quantization with graph structures [11]. Binary hashing methods such as ITQ [10] similarly rely on distributional alignment. Across these lines, the standard assumption is that query and database statistics are in-distribution and approximately centroid-aligned.

Table 1: BQ methods (R@100, Vanilla / PMC) on MSCOCO (CLIP-B/32) and AudioCaps (IB). RotatedBinary = rotated BinaryFlat (BBQ-style).

	MSCOCO		AudioCaps	
Method	t⟶i	i⟶t	t⟶a	a⟶t
BinaryFlat	.51 / **.57**	.46 / **.57**	.63 / **.66**	.51 / **.64**
BinaryIVF	.51 / **.57**	.47 / **.57**	.59 / **.61**	.50 / **.63**
RotatedBinary	.29 / **.58**	.21 / **.57**	.65 / **.70**	.63 / **.69**
RaBitQ	.58 / **.64**	.52 / **.61**	**.75** / **.75**	.67 / **.75**

Algorithm 1: PMC Index Construction

Require: Database $\mathcal{X}$ (modality b), query sample Q (modality a), interpolation α
Ensure: IVF-based quantized index $\mathcal{I}$ and gap $\mathbf{g}$

$\boldsymbol{\mu}_q \leftarrow \mathrm{mean}(Q)$; $\boldsymbol{\mu}_x \leftarrow \mathrm{mean}(\mathcal{X})$
$\mathbf{g} \leftarrow \boldsymbol{\mu}_q - \boldsymbol{\mu}_x$
for each $\mathbf{x} \in \mathcal{X}$ **do**
  $\mathbf{x}' \leftarrow \mathrm{normalize}(\mathbf{x} + \alpha\,\mathbf{g})$
end for
$\mathcal{I} \leftarrow \mathrm{BuildIndex}(\{\mathbf{x}'\})$ ▷ RaBitQ, BBQ, BinaryFlat
return $\mathcal{I}$, $\mathbf{g}$

Cross-modal and OOD retrieval. Cross-modal and OOD retrieval methods usually intervene outside compressed code formation. OOD-DiskANN [13, 27] augments graph search for distribution shift; RoarGraph/DEG [2, 32] modify graph connectivity; DeDrift [1] and TCR [20] use test-time adaptation or retraining-style updates to counter drift. They are effective but usually assume uncompressed vectors, additional graph structures, or online serving updates. PMC is complementary, not a direct baseline: one offline centroid correction in the 1-bit compressed regime, with no added index-time memory or retraining.

Modality gap. Mind-the-Gap [21] shows that cross-modal embedding spaces often exhibit stable centroid offsets between modalities. Follow-up analyses attribute this behavior to training dynamics [25], characterize its geometry [18], and study mean-subtraction effects [35]. Other work either preserves the gap [24] or reduces it through encoder retraining [5]; see also the recent survey [31]. No prior work addresses how the modality gap degrades *compressed* ANN indexes, where centroid misalignment corrupts both IVF routing and quantized codes; this is our focus.

A closely related line, gap-aware calibration for mixed-modality ranking [19], rescales scores to offset the centroid gap but works on uncompressed vectors, never enters code formation, and still subtracts the query mean at query time; PMC instead folds the correction into the database index at α=1, rewriting IVF routing and binary codes at the source with no serving-time query transform.

3 Method

3.1 Problem Setup

Let ϕ be a frozen multimodal encoder mapping inputs from modalities a and b into $\mathbb{R}^d$. The database $\mathcal{X} = \{\phi(x_i)\}$ consists of modality-b embeddings; queries $q = \phi(q_j)$ come from modality a. We denote the per-modality means $\boldsymbol{\mu}_q = \mathbb{E}[\phi(q)]$ and $\boldsymbol{\mu}_x = \mathbb{E}[\phi(x)]$, and the *modality gap* $\mathbf{g} = \boldsymbol{\mu}_q - \boldsymbol{\mu}_x$. The goal is to build a compressed ANN index over $\mathcal{X}$ that answers cross-modal queries with high recall under a fixed memory budget.

Table 2: PMC main retrieval results (R@100, α=1, IVFRaBitQFastScan) at the two operating points of Sec. 4.1: *No reranking* and *With reranking*. Each cell Vanilla / MeanShift / PMC; bold = best per cell.

			No reranking			With reranking (K'=400)		
Dataset	**Enc.**	$\lVert g \rVert$	n_{probe}	$q{\to}db$	$db{\to}q$	n_{probe}	$q{\to}db$	$db{\to}q$
MSCOCO	CLIP	.82	16	.58 / .54 / **.63**	.50 / .49 / **.60**	64	.81 / .79 / **.90**	.72 / .74 / **.87**
	CL-L	.82	16	.55 / .53 / **.65**	.47 / .51 / **.63**	64	.79 / .77 / **.91**	.66 / .73 / **.89**
	IB	.70	16	.67 / .62 / **.75**	.71 / .66 / **.75**	64	.93 / .87 / **.97**	.93 / .88 / **.95**
Flickr30K	CL-L	.77	32	.41 / .34 / **.48**	.33 / .37 / **.48**	32	.91 / .85 / **.97**	.81 / .82 / **.96**
Clotho	IB	.61	16	.72 / .60 / **.73**	.62 / .51 / **.69**	32/64	**1.00** / .96 / **1.00**	.92 / .87 / **.98**
AudioCaps	IB	.61	8/16	.75 / **.78** / **.78**	**.83** / **.83** / **.83**	64	**.98** / .88 / **.98**	.90 / .84 / **.97**
LAION-400M	CLIP	.72	256	.108 / .074 / **.143**	.069 / .043 / **.073**	256	.198 / .149 / **.277**	**.158** / .080 / .140

$q{\to}db$: text-to-image/audio; $db{\to}q$: reverse. CL-L = CLIP-L; IB = ImageBind; $\lVert g \rVert$ = modality-gap norm; MeanShift = query-only mean shift. n_{probe} split $q{\to}db/db{\to}q$ where directions differ (AudioCaps, Clotho).

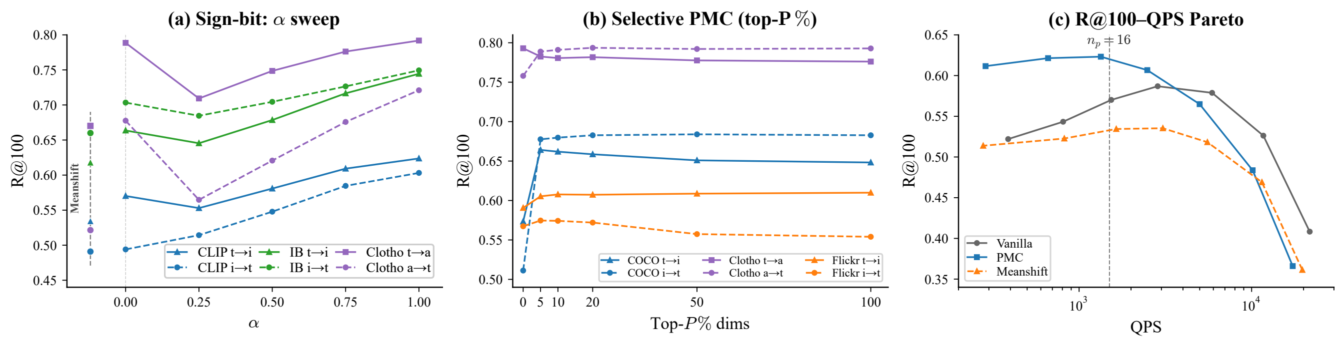

Figure 3: (a) BQ R@100 vs. shift strength α on MSCOCO+Clotho (α=0: MeanShift, α=1: full DB-side PMC): α=1 is best or near-best, justifying the default. (b) Selective PMC: on concentrated-gap MSCOCO, top-P=5% already reaches peak R@100; diffuse gaps require broader correction. (c) R@100–QPS Pareto: PMC dominates Vanilla across throughput levels.

3.2 BQ Vulnerability to Modality Gap

BQ encodes each dimension as $\hat{x}_i = \text{sign}(x_i - c_i)$ relative to centroid c, as in RaBitQ [7] and BBQ (Apache Lucene). With one bit, the decision boundary passes exactly through c with *zero error margin.* Let $\delta_i = x_i - c_i$ be the residual under $c = \boldsymbol{\mu}_x$ (so $b_i = \text{sign}\,\delta_i$); shifting to $c^* = \boldsymbol{\mu}_x + \alpha\mathbf{g}$ flips dimension i's sign iff

$$\delta_i(\delta_i - \alpha g_i) < 0\,, \quad \text{i.e.,} \quad 0 < \frac{\delta_i}{\alpha g_i} < 1\,. \tag{1}$$

Under smooth symmetric residual densities p_i near the decision boundary, a first-order expansion yields the expected number of flips:

$$\mathbb{E}[F] \;\approx\; \alpha \sum_i p_i(0)\,|g_i|\,, \tag{2}$$

a density-weighted ℓ_1 norm of the gap. Eq. (1) shows that flip risk in dimension i grows with $|\alpha g_i|/|\delta_i|$. When gap energy is concentrated—in CLIP-L on MSCOCO, top 10% of dimensions carry ≈90% of $\lVert \mathbf{g} \rVert^2$—corruption is *structured* rather than random, biasing Hamming distance estimates systematically. The same offset misaligns IVF cluster assignments, compounding routing errors with code corruption. Eq. (2) formalizes this scaling, explaining the gap-proportional gains in Table 2 and the top-5% selective PMC result on concentrated-gap encoders (Figure 3b). Multi-bit quantization (IVFPQ [14], OPQ [8]) partially absorbs displacement through wider error margins; BQ is maximally vulnerable and is the focus of our main claims (multi-bit: Section 4.5). PMC applies to all sign$(\cdot - c)$ methods (BinaryFlat, BinaryIVF, BBQ, RaBitQ) and analogous schemes such as LSH [29].

3.3 PMC Correction

PMC applies a one-time centroid alignment *before* index construction. From a training sample of modality-a queries and the modality-b database, we compute the gap $\mathbf{g}$ and shift database vectors toward the query centroid:

$$\mathbf{x}' = \frac{\mathbf{x} + \alpha\,\mathbf{g}}{\lVert \mathbf{x} + \alpha\,\mathbf{g} \rVert}\,, \qquad \mathbf{q}' = \frac{\mathbf{q} - (1-\alpha)\,\mathbf{g}}{\lVert \mathbf{q} - (1-\alpha)\,\mathbf{g} \rVert} \tag{3}$$

where $\alpha \in [0,1]$ interpolates between full query-side correction (α=0, equivalent to mean shift) and full database-side alignment (α=1). The BQ index is then built on $\{\mathbf{x}'\}$; in our main setting α=1 (full database-side alignment), query-time transformation is identity ($\mathbf{q}'$=$\mathbf{q}$), so serving incurs no extra cost.

Effectiveness of $\boldsymbol{\alpha}$=1. Per-vector renormalization after the shift preserves unit-norm structure; the α-sweep in Figure 3a confirms α=1 as best or near-best in all settings.

Selective PMC as a mechanism test. We restrict correction to top-P% dimensions ranked by $|g_i|$:

$$g_{\text{sel},i} = \begin{cases} g_i & \text{if } |g_i| \ge |g|_{(P)} \\ 0 & \text{otherwise} \end{cases}\,, \qquad \mathbf{x}'_{\text{sel}} = \frac{\mathbf{x} + \alpha\,\mathbf{g}_{\text{sel}}}{\lVert \mathbf{x} + \alpha\,\mathbf{g}_{\text{sel}} \rVert} \tag{4}$$

where $|g|_{(P)}$ is the top-P% magnitude threshold on $|g_i|$. The cumulative energy captured by $\mathbf{g}_{\text{sel}}$ is

$$\mathcal{E}(P) = \frac{\sum_{i \in \mathcal{S}_P} g_i^2}{\lVert \mathbf{g} \rVert^2}\,, \quad \mathcal{S}_P = \left\{ i : |g_i| \ge |g|_{(P)} \right\}. \tag{5}$$

Since flip risk concentrates in high-$|g_i|$ dimensions (Eq. (1)), correcting only the highest-energy ones suffices: for CLIP the top 10%

Table 3: R@10 at the no-reranking operating points of Table 2. Each cell Vanilla / MeanShift / PMC; bold = best per cell.

Dataset	Enc.	$q \to db$	$db \to q$
MSCOCO	CLIP	.40 / .38 / **.46**	.29 / .30 / **.39**
	CL-L	.36 / .37 / **.48**	.26 / .35 / **.44**
	IB	.55 / .51 / **.63**	.57 / .54 / **.64**
Flickr30K	CL-L	.31 / .26 / **.38**	.22 / .29 / **.38**
Clotho	IB	.59 / .45 / **.60**	.48 / .35 / **.54**
AudioCaps	IB	.39 / .43 / **.44**	.44 / .46 / **.48**
LAION-400M	CLIP	.075 / .038 / **.086**	.035 / .028 / **.048**

Table 4: Mechanism/control checks (α=1). J@100: top-100 Jaccard vs. exact-IP GT; cos@25: calibration cosine at n_{calib}=25.

Direction	Flip%	J@100	BinaryFlat R@100	cos@25
MSCOCO t→i	16.14	.585	.507 → .570	.986
MSCOCO i→t	15.99	.542	.458 → .568	.986
AudioCaps t→a	14.34	.801	.630 → .660	.971
AudioCaps a→t	17.84	.799	.514 → .644	.955

by $|g_i|$ carry 86–92% of total gap energy, so P=5% already recovers peak recall; ImageBind's diffuse gap (≈72%) requires full-vector correction.

4 Experiments

4.1 Setup

We evaluate three frozen backbones—CLIP ViT-B/32 (d=512) [23], CLIP ViT-L/14 (d=768), ImageBind (d=1024) [9]—on image–text and audio–text retrieval: MSCOCO 5K [22], Flickr30K (n=31K) [33], AudioCaps [17] (884 clips, 4,415 captions; audio DB), Clotho v2 (n=5,926) [4], and LAION-400M (407M vectors, 410 shards; exact inner-product (IP) ground truth for 10K random queries per direction via per-shard IndexFlatIP merged into a global top-100—a full-corpus brute-force scan) [26]. Indexes use FAISS [3, 16]: IVFRaBitQFastScan (1-bit, 33–136 B/vec) and IVFPQ/OPQ [8, 14] (m=16, 64 B/vec); per FAISS's IVF heuristic, $n_{\text{list}}=\lceil\sqrt{n}\,\rceil$ and $n_{\text{probe}}\approx n_{\text{list}}/4$, matched across methods; with reranking (Sec. 4.3) we set $n_{\text{probe}}\approx n_{\text{list}}$ and exact-rerank the top K'=400; LAION uses n_{list}=80K≈$4\sqrt{N}$, n_{probe}=256, 30.4 GB at 72 B/vec (28×); PMC leaves bytes/vector identical to Vanilla; BQ baselines: BinaryFlat, BinaryIVF, rotated BinaryFlat (BBQ-style). We report R@100 against exact-IP ground truth on original (unshifted) embeddings, except AudioCaps, which uses standard caption annotations (one clip per caption; that clip's captions in reverse)—PMC modifies only the index-internal representation (analogous to OPQ rotation), so gains reflect better approximation, not redefined relevance—and QPS (queries per second) as the single-threaded median over 5 runs. All runs use a single Intel Core i7-12700F, 32 GB RAM, CPU-only.

4.2 Main Results

PMC improves RaBitQ recall in nearly all tested configurations, with gains scaling with gap magnitude (Table 2): Flickr30K i→t reaches +45% and MSCOCO CLIP-L i→t +34%. MeanShift can hurt—it leaves IVF centroids misaligned, degrading routing—while PMC improves or matches both directions. Across all 16 BQ method × direction configurations, PMC improves or matches R@100 without exception (Table 1); the same ordering holds at R@10, where PMC is best in all 14 dataset × direction configurations (Table 3), so the gain is not confined to deep result lists. RotatedBinary suffers most—rotation spreads the concentrated gap energy across all bits—and gains most from PMC. PMC adds no per-query work, so QPS tracks Vanilla across all n_{probe} (Figure 3c).

$\|\mathbf{g}\|$ as predictor. The ratio condition (1) implies more flips at higher gap, and R@100 gains track $\|\mathbf{g}\|$: MSCOCO CL-L ($\|\mathbf{g}\|$=.82) and Flickr30K CL-L ($\|\mathbf{g}\|$=.77) show the largest gains, whereas AudioCaps ($\|\mathbf{g}\|$=.61) gains only marginally—its gap is small relative to intra-class variance. Direct bit-level checks corroborate this (Table 4): PMC induces 14–18% sign-bit flips while preserving nontrivial ground-truth top-100 overlap (J@100=0.54–0.80) and repairs BinaryFlat R@100 in every direction.

At 407M-vector deployment scale, LAION-400M ($\|\mathbf{g}\|$=.72) confirms the gap-proportional pattern in both directions (Table 2).

Table 5: MSCOCO/CLIP-B/32: component ablation and IVF-RaBitQ controls. Rand/Shuf are mean±std over 5 seeds.

Component ablation Index	Dir.	Van.	Q-only	DB-only	Both
BinaryFlat	t→i	.507	.569	**.570**	**.570**
BinaryFlat	i→t	.458	**.571**	.568	.569
IVF-RaBitQ	t→i	.578	.541	**.637**	.599
IVF-RaBitQ	i→t	.515	.495	**.608**	.554

IVF-RaBitQ controls Dir.	Van.	DB-PMC	Rand	Shuf	S-Flip	NoNorm
t→i	.578	**.637**	.562±.009	.552±.006	.514	.555
i→t	.515	**.608**	.564±.009	.559±.009	.491	.520

Table 6: PMC generality on IVFPQ and OPQ across encoders (R@100 Vanilla / PMC, mean over both directions; per-direction omitted).

Dataset	Enc.	IVFPQ		OPQ	
		R@100	Δ	R@100	Δ
MSCOCO	CLIP	.52 / **.64**	+23%	.64 / **.67**	+4%
	CL-L	.39 / **.61**	+56%	.57 / **.67**	+17%
	IB	.59 / **.69**	+17%	.70 / **.77**	+9%
Flickr	CL-L	.47 / **.50**	+7%	.51 / **.52**	+1%

4.3 Robustness under Exact Reranking

With reranking [30] we exact-rescore the top-K' candidates (Sec. 4.1; identical per-method budget). Vanilla's no-rerank recall is capped by what its corrupted codes admit; PMC repairs the pool and carries this lead into reranking, improving or matching Vanilla and MeanShift in every cell except LAION-400M reverse across $K' \in \{100, 200, 400, 500\}$ (MSCOCO t→i PMC .62/.80/.90/.93 vs. Vanilla .53/.69/.81/.85; Table 2 reports K'=400), reaching $R@100 \geq 0.90$ at a smaller budget. There, exact rescoring lets Vanilla overtake PMC at $K' \geq 200$—a scan-depth effect: LAION probes only 0.32% of its lists (n_{probe}=256), so PMC's flatter pool covers fewer true positives, whereas small corpora rerank near-exhaustively.

4.4 Ablation Study

Calibration is stable from n_{calib}=25 (cosine 0.955–0.986 across modalities, Table 4; R@100 flat within .003 up to n_{calib}=400), making PMC practical with few cross-modal samples. BinaryFlat is insensitive to shift direction, but IVF-RaBitQ strongly prefers DB-only correction (.637/.608) over Q-only (.541/.495) or both sides (.599/.554)—shifting both re-displaces the routing pivot, confirming the centroid must be corrected at the source, where it is both routing pivot and code boundary. Four controls (random, shuffled, sign-flipped, unnormalized shift) all trail DB-PMC (Table 5), ruling out arbitrary perturbation.

4.5 Multi-Bit Generality

PMC generalizes beyond 1-bit: averaged over both directions it improves IVFPQ and OPQ across all encoders and datasets (audio omitted; n_{db} too small for codebook training), with the gap-proportional trend persisting (Table 6). OPQ's learned rotation partly absorbs the shift, giving smaller but complementary gains; IVFPQ lacks this and benefits more.

5 Conclusion

PMC corrects cross-modal centroid mismatch in compressed ANN with a build-time, database-side shift that rewrites routing pivots and sign-bit boundaries at the source, yielding gains that scale with modality-gap magnitude at no extra index memory and, at α=1, no serving-time query transform. The one regime favoring uncorrected codes—deep reranking under sub-1% list probing at 400M scale—is a scan-depth effect. Drift-adaptive α, graph-based ANN, and billion-scale deployment remain future work.

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GenAI Usage Disclosure

Generative AI tools (Claude) were used for code scaffolding and LaTeX formatting. All experimental design, results, analyses, and scientific claims were independently produced and verified by the authors.

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